

# Richy Koshy

Python engineer: GenAI, backend and automation

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**5+ years of Python development**, plus **a year of industrial automation**. Shipped production GenAI, backend, and automation systems for a US mortgage originator: an enterprise Azure RAG platform, a custom Encompass REST API SDK, and 13 production automations across compliance and investor operations.

## Work experience

### Software Developer

#### Primary Residential Mortgage, Inc.

US mortgage originator (PRMI). Owned requirements, architecture, build, UAT and production. Sole engineer on the RPA automations; full technical ownership of MAX AI.

- **MAX AI, enterprise RAG platform.** Owned the full Python backend across three Azure services; now PRMI's primary knowledge-retrieval tool **organization-wide**.
  - **Retrieval:** hybrid vector + semantic ranking on Azure AI Search with HyDE query expansion; Azure OpenAI GPT-4o with ada-002 embeddings; one index per project, with multi-tenant access mapping in SQL Server behind Azure AD auth.
  - **Ingestion:** Durable Functions pipeline, idempotent via audit-table checkpointing with per-file failure isolation; local PDF partitioning and table extraction with unstructured (poppler + Tesseract), shipped as a Docker image on Azure Container Registry; Azure Document Intelligence for scans.
  - **Chat:** FastAPI service streaming NDJSON with metadata-first framing, so citations render before the first token.

Read the MAX AI deep dive ([richie-rk.github.io/max-ai.html](https://richie-rk.github.io/max-ai.html))

- **Final Docs refactor.** Rebuilt a monolithic automation into config-driven dynamic dispatch: one YAML config per investor and a dispatch table that maps each investor to its handler functions, treating each investor as data, not code. **4 new investors onboarded in 1 month** vs. a full quarter before; onboarding cut from **2-4 weeks to days**; adopted as the template for 2 other initiatives.

Read the Final Docs deep dive ([richie-rk.github.io/final-docs.html](https://richie-rk.github.io/final-docs.html))

“PRMI now treats investor expansion as a business decision, not a dev-capacity constraint.”

Corban Wells, VP of Technology Product Management, PRMI  
Read Corban Wells's full letter ([richie-rk.github.io/letter.html](https://richie-rk.github.io/letter.html))

- **Encompass REST API SDK.** Internal Python package consumed by all 13 automations: OAuth2 token caching, concurrent paginated pipeline reads (ThreadPoolExecutor), transactional loan writes with a lock → write → read-back → unlock pattern that prevents lost updates on shared loans, eFolder document CRUD, disclosure tracking, and 429 retry with backoff.
- **13 production RPA automations** (Power Automate + Python) across ECOA adverse action, TRID, HMDA, MERS sync, final docs, HFA exceptions, Blend credit management, warehouse collateral, and suspense resolution across 9 investors; plus 3 OCR pipelines (Tesseract + Azure Form Recognizer) extracting credit data from PDF reports.
- **Delivery ownership.** Led BRD/PRD gathering, flow diagrams, first-call sign-offs, UAT coordination, and documentation for every delivered automation.

## Software Developer

### Antz AI Solutions

Client: TechSpan Engineering (IoT)

- **HydroQ API.** Designed the FastAPI interface for the database CRUD operations of a multi-tenant remote water-quality monitoring platform: SQLAlchemy with a generic CRUD layer over PostgreSQL and MSSQL; replaced **2-4 hours of manual onboarding per site** with API-driven workflows; full endpoint test coverage.

## Industrial Automation Engineer

### Goodyear South Asia Tyres

Tyre manufacturing: PLCs, drives and controls

- Programmed and maintained PLCs and variable-frequency drives across tyre-curing and material-handling lines.
- Led breakdown troubleshooting on production-critical electrical systems, minimizing line downtime.
- Commissioned CTMS & BMS systems alongside Rockwell and Honeywell engineers.

## Skills and tools

### GenAI and RAG

Platforms                      Azure OpenAI, Azure AI Search, LangChain

Techniques                    Hybrid retrieval, HyDE, semantic ranking, RAG evaluation

### Backend

Python                         FastAPI, SQLAlchemy, Pydantic, asyncio, pytest

APIs                            REST design, OAuth2, NDJSON streaming, microservices

### Cloud and infrastructure

Azure                          Functions, Durable Functions, Container Registry, Key Vault, Document Intelligence

Tooling                        Docker, Git, SQL Server, PostgreSQL

### Automation and domain

RPA                            Power Automate, Encompass API, OCR (Tesseract)

Domain                        US mortgage: ECOA, TRID, HMDA, MERS, investor ops

## GitHub repositories

- rag-app-azure (<https://github.com/richie-rk/rag-app-azure>): full-stack Azure RAG
- EncompassRest-RAG-Assistant (<https://github.com/richie-rk/EncompassRest-RAG-Assistant>): hybrid FAISS + BM25
- PyTeslaCoil (<https://github.com/richie-rk/PyTeslaCoil>): physics calculator, 73 tests

## Education and awards

BE Electrical Engineering

University of Mumbai

- **Best Paper**, Intl. Conference on Advanced Trends in Engineering
- **1st Place**, Tantragyan National Project Competition

## Reference

Corban Wells, VP of Technology Product Management, PRMI. Letter available on request. Read Corban Wells's full letter ([richie-rk.github.io/letter.html](https://richie-rk.github.io/letter.html))